

Sofia Loya

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PROFESSIONAL SUMMARY

Mechanical Engineer and Carnegie Mellon M.S. candidate with experience spanning medical devices, electronics, robotics, and AI-enabled product development. Four years delivering high-impact mechanical designs, test automation, and product reliability through FEA, material selection, and design verification in regulated, high-volume environments.

EDUCATION

Carnegie Mellon University Master of Science in Mechanical Engineering GPA: 4.0/4.0 Rales Fellows Program • Selected Coursework: Machine Learning and Artificial Intelligence for Engineers, Introduction to Deep Learning	Pittsburgh, PA May 2027
Texas A&M University Bachelor of Science in Mechanical Engineering GPA: 3.587/4.0	College Station, TX May 2021

PROFESSIONAL EXPERIENCE

Medtronic <i>Quality Engineering Intern / Cardiac Ablation Solutions</i> • Drove EU Clinical Study compliance by developing and evaluating seven test methods and characterization tests for a small focal catheter • Automated measurement collection on LCR Meter for capacitive coupling test method that reduced 99 manual data entry points to one data file through a python script	Mounds View, MN June 2026 – Aug 2026
Dell Technologies <i>Mechanical Engineer II / Dell Commercial Mobile Latitude</i> • Delivered \$500M in cost avoidance by redesigning the LCD assembly, driven by adhesive performance evaluation, mechanical stress analysis, and fatigue testing to develop internal design specifications • Conducted ANSYS simulations comparing injection-molded hinge designs to identify geometry improvements that reduce stress concentrations and guide future design standards • Reduced weight of a high-volume platform via mechanical redesign in CAD and material selection, decreasing weight of single platform by 10% while maintaining stiffness and reliability targets • Reviewed 8 bills of materials with 200+ parts each and drove design changes to reduce costs by 12M USD • Diagnosed eDP cable design inconsistencies across platforms and partnered with design, electrical, and manufacturing teams to simplify configurations, achieving a 15% reduction in part complexity	Austin, TX April 2023 – June 2025
Dell Technologies <i>Mechanical Engineer I / Dell Commercial Mobile Rugged</i> • Developed and filed patent for electromechanical intrusion-prevention design integrating precision sensor and secure mechanical engagement to protect customer data • Built I/O port housing in CREO and analyzed in ANSYS, reducing testing time by one week • Created a Power BI dashboard comparing mobile platforms and competitors, enabling faster executive decision-making • Led a cross-functional team to design and pitch a growth framework aimed at scaling a \$240M security business to \$2B	Austin, TX June 2021 – April 2023
Dell Technologies <i>Mechanical Engineer Intern / Dell Desktop Precision</i> • Prototyped housing for removable SSD cards in Creo for Precision tower quadrupling memory capacity • Collaborated with engineering team to design module that increases capacity of Precision towers by 8 TB	Austin, TX May 2020 – August 2020
Ethicon <i>R&D Intern / Laparoscopic Surgical Products</i> • Designed and built a test fixture for laparoscopic surgery simulation enabling realistic use of laparoscopic devices	Cincinnati, OH January 2019 – May 2019

PROJECTS

Insulator Defect Identification <i>Deep Learning – YOLOv8, Ultralytics, PyTorch</i> • Wrote and trained YOLOv8 object-detection model to identify defective power-line insulators in aerial drone imagery, achieving 99.5% mAP@50 on string detection and 92.8% mAP@50 on shell-defect classification
Bio-Inspired Robot for Multi-modal Locomotion • Performed motor torque analysis for a 7-module bio-inspired snake robot to inform actuator selection • Executed physical launch testing and statistical analysis (Anderson-Darling normality tests, two-sample t-tests) across 20 trials, demonstrating a statistically significant 8.7% increase in glide distance with undulation at 1.8 m/s ($p = 0.010$)

SKILLS

CAD, Simulation & Hardware: SolidWorks, Creo, Finite Element Analysis, Gazebo, Mujoco, Arduino, LabVIEW, rapid prototyping
Programming: Python, MATLAB, ROS 2; Machine Learning: PyTorch, OpenCV, Ultralytics, Hugging Face Transformers
Data Analytics: applied statistics, Power BI, reliability testing, JIRA, Agile